DEPARTMENT OF MATHEMATICS, IIT MADRAS

MA5950 — Mathematical Finance: Theory & Implementation

Instructor: Prof. Neelesh Upadhye

Lecture 18: Pricing American vs. European Options

Early Exercise · The Exercise Boundary · Why Calls Are Different

September 9, 2026

Today's Agenda

1. The right to act early

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3. A worked example: the American put
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5. Put–call parity

Learning Objectives

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- Price an American option by modified backward induction.
- Identify early-exercise nodes and compute the premium.
- Explain why American calls (no dividends) are never exercised early.

A Fixed Appointment vs. an Open Ticket

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The Dominance Argument

Key Takeaway

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An American holder can always replicate the European strategy. More choices can't hurt:

$$V^{\text{American}} \geq V^{\text{European}}$$

The gap is the **early exercise premium**.

The American Recursion

Key Takeaway

$$V_{n,j} = \max\left\{ \underbrace{\text{payoff}(S_{n,j})}_{\text{exercise now}},\ \underbrace{\frac{1}{1+r}\left[qV_{n+1,j} + (1-q)V_{n+1,j+1}\right]}_{\text{continuation}} \right\}$$

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The *only* change is the outer max.

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The American Put: Setting Up

S_0=100, u=1.2, d=0.9, r=5%, K=105, N=2, q=0.5

Lattice: $\{144, 108, 81\}$ at t=2; $\{120, 90\}$ at t=1.

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Terminal payoffs $\max(K - S, 0)$: $V_{2,0}$=0, $V_{2,1}$=0, $V_{2,2}$=24.

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$15 > 11.4286 \Rightarrow$ EXERCISE EARLY

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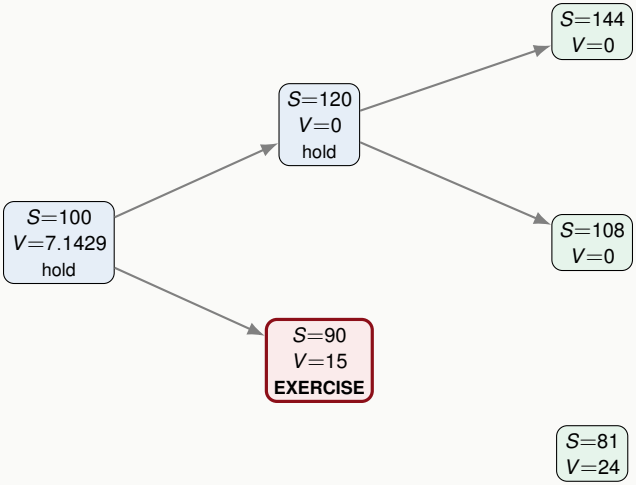
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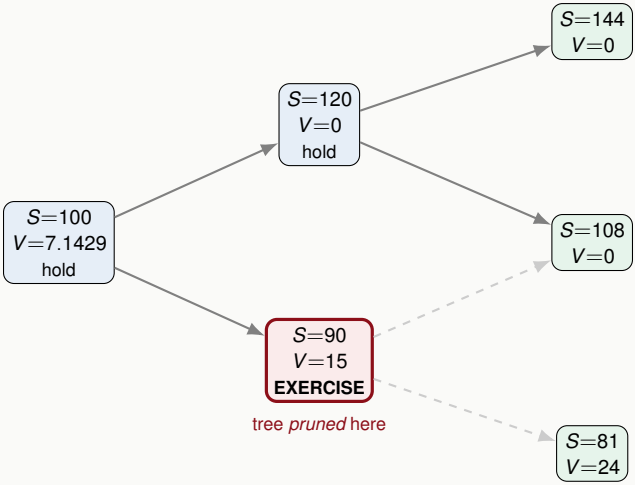
European put $= 5.4422$. American put $= 7.1429$.

early exercise premium $= 1.7007$

The Pruned Tree



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Exercising hands you Rs. 15 *now*.

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The Payoff Is Capped

Exercising hands you Rs. 15 *now*. Holding gets a risk-neutral expected Rs. 12, arriving *later* – worth Rs. 11.43 today. The key: a put's payoff is **capped** at K. Once the stock is low, most of that ceiling is reached – waiting mainly costs you time value on cash you could hold now.

No Early Exercise for American Calls

Key Takeaway

For a call on a **non-dividend-paying** stock, early exercise is *never* strictly optimal:

$$V^{\text{American call}} = V^{\text{European call}}$$

Early exercise premium $= 0$.

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t=1 up: continuation 20 $>$ exercise 15.

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t=1 up: continuation 20 $>$ exercise 15. t=0: continuation 10.2041 $>$ exercise 0. Never exercise. American $=$ European $=$ 10.2041.

Two Reasons Never to Exercise a Call Early

Both Point the Same Way: Wait

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Unlike a put, a call's payoff is **unbounded above** – no "ceiling reached" effect.

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A large dividend causes the stock to drop on the ex-dividend date – and a call holder receives none of it. Exercising *just before* the drop, to capture the dividend as a shareholder, can then be optimal. This is why early exercise of American calls is observed almost exclusively before large dividends.

Put–Call Parity (European Only)

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$$C_0 - P_0 = S_0 - \frac{K}{(1+r)^N}$$

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$C_0 - P_0 = 10.2041 - 5.4422 = 4.7619$

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Checking on our tree

$C_0 - P_0 = 10.2041 - 5.4422 = 4.7619$
$S_0 - K/(1.05)^2 = 100 - 95.2381 = 4.7619$ ✓

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Parity compares two portfolios with *identical, fixed-date* payoffs. American options can be exercised at different, decision-dependent times. The portfolios no longer match state-by-state – the equality degrades into inequalities. Always check which version you're invoking.

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- Our example: European put 5.4422, American put 7.1429, premium 1.7007.
- Calls on non-dividend stocks: **never** exercise early, so American = European.
- Put–call parity holds for **European** options only.

Check Your Understanding

Discuss (2 minutes)

1. $S_0=50$, $u=1.1$, $d=0.9$, $r=4\%$, $K=52$, $N=2$: price European and American puts. Premium?

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2. Why does a put's *capped* payoff invite early exercise but a call's *unbounded* payoff not?
3. A call is exercised early before a big dividend. Which "never exercise" argument is overridden?

Looking Ahead

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Brownian Motion – Week 7

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Brownian Motion – Week 7

The destination the convergence plot was already pointing toward.